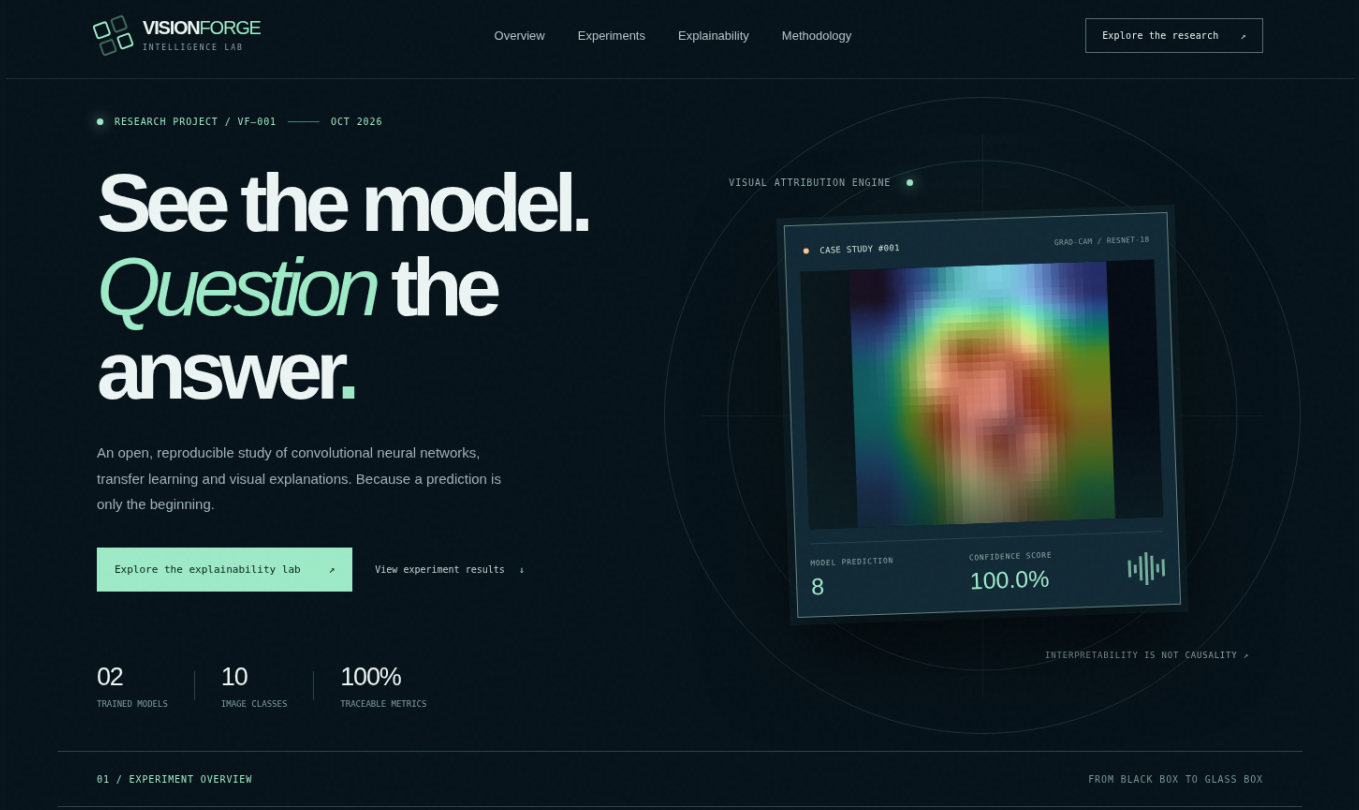
◈ **VISIONFORGE**

PERCEPTION / EXPLANATION / REPRODUCIBILITY

See the model.
Question the answer.

A controlled comparison of a trained custom CNN and a rotation-pretrained ResNet-18 with real Grad-CAM explanations.



RESEARCH REPORT / VF-001 / OCTOBER 2026

Executed PyTorch CPU benchmark • 2 saved models • 315 held-out test images

Independent project. Not affiliated with or endorsed by BTU Cottbus-Senftenberg.

01 / RESEARCH DESIGN

The experiment at a glance

Hypothesis-driven model comparison with fixed partitions

Research question

Can a compact convolutional neural network match or outperform a larger ResNet-18 with within-domain rotation pretraining on small image data? Can both approaches be examined using class-specific visual attribution?

DATASET	TRAIN	VALIDATION	TEST
1,797 images	**1,168 images**	**314 images**	**315 images**
scikit-learn handwritten digits	supervised + rotation pretext	best epoch selection	holdout; same for both models

Training routes

MODEL A / SUPERVISED CNN

Learned 32/64/96 channel convolution stages; random initialization. AdamW, learning rate 0.002, 14 epochs; validation-best at epoch 11.

MODEL B / RESNET-18 TRANSFER

Pretrain a 1-channel ResNet-18 to recognize 0/90/180/270-degree image rotations using training images only. Replace the head with 10 outputs and fine-tune all layers for 6 epochs.

COMMON EVALUATION

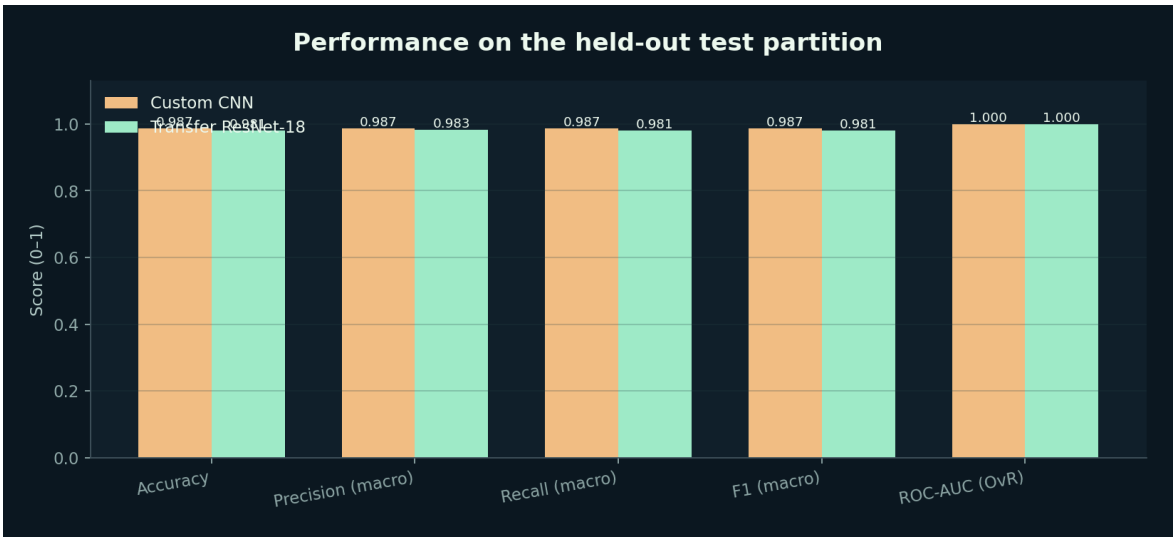
Predict exactly once on the same 315-image holdout after validation selection. Report classwise precision/recall/F1, macro OvR ROC-AUC, ECE and bootstrap intervals.

02 / HELD-OUT RESULTS

What the models actually achieved

Measured on the same 315 test images; numbers from saved predictions

METRIC	CUSTOM CNN	TRANSFER RESNET-18
Accuracy	98.730%	98.095%
Macro precision	98.749%	98.286%
Macro recall	98.729%	98.083%
Macro F1	98.724%	98.120%
Macro OvR ROC-AUC	0.999887	0.999932
10-bin ECE (lower better)	0.044678	0.009299
Correct images	311 / 315	309 / 315

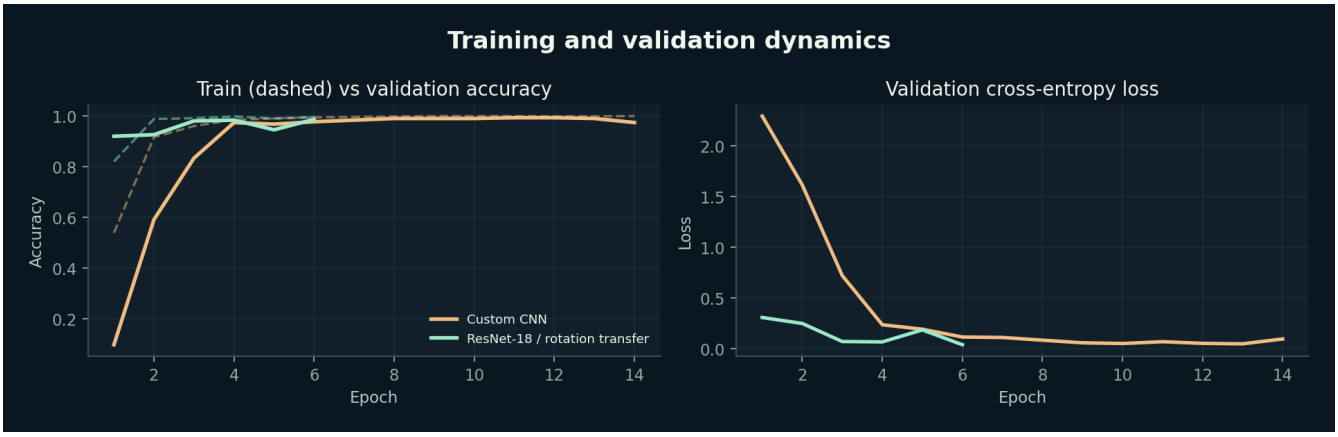


Interpretation

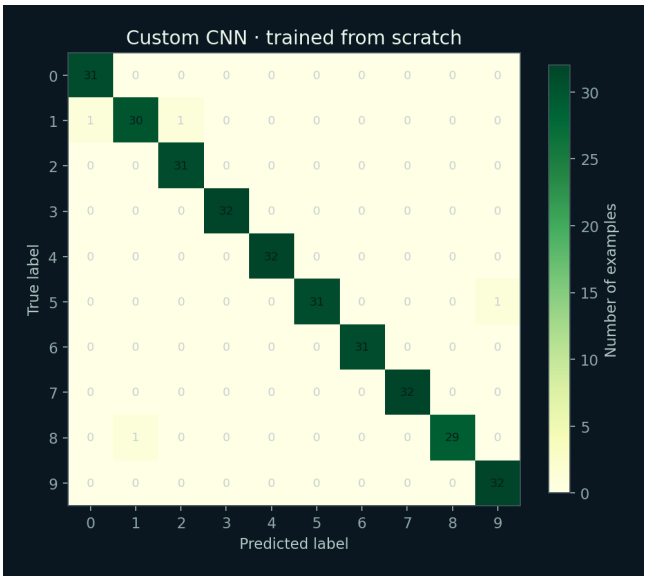
The smaller CNN is slightly more accurate (two additional correct test images). The rotation-pretrained ResNet-18 has marginally higher OvR ROC-AUC and a smaller 10-bin ECE. These differences are exploratory: one random split, one seed and unequal training budgets cannot establish a statistically significant architecture advantage.

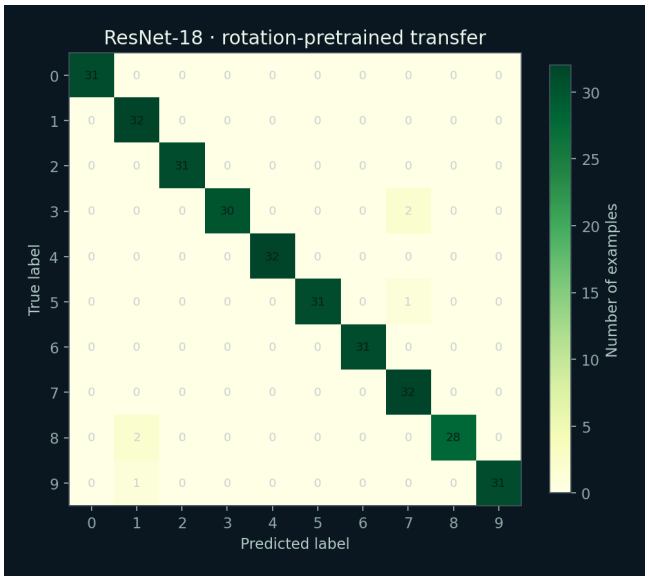
Learning, errors and uncertainty

Best validation checkpoint selected before held-out evaluation



Confusion matrices - 10 digits, 315 samples

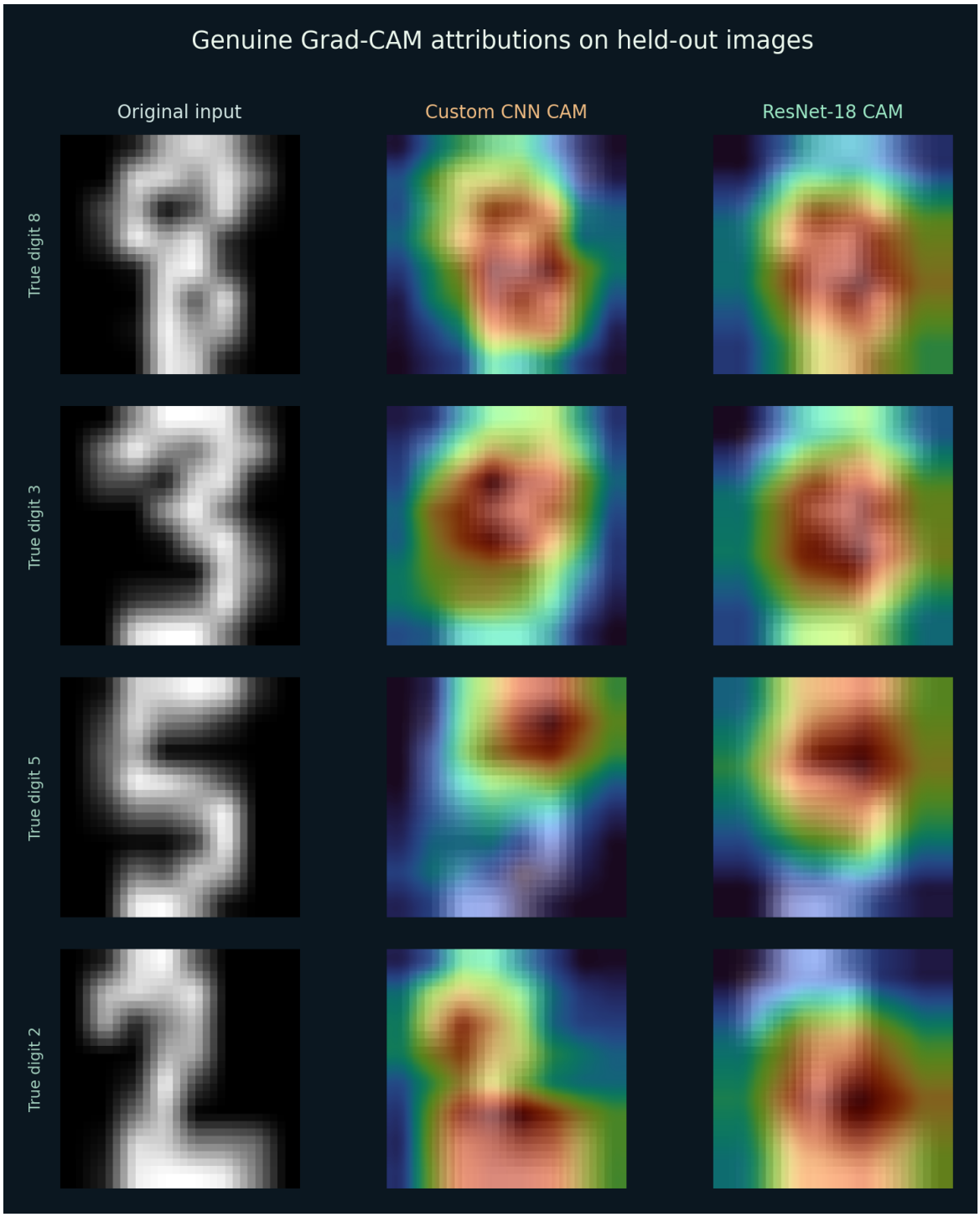




See additional calibration and ROC curves under docs/assets/. Confusion matrix cells are true/predicted class counts. The bootstrap 95% accuracy intervals are 97.46-99.68% for CNN and 96.51-99.37% for ResNet-18.

Class-specific Grad-CAM

Maps generated from actual gradients of saved checkpoints on held-out samples



The heatmap indicates class-sensitive feature regions. It is not a segmentation mask, proof of causality, or a reliability guarantee. The web workbench allows the reader to switch between both models and inspect 12 held-out cases.

A research workflow, not a claim

Curriculum relevance and project limitations

Package contents

- Complete Python/PyTorch source, two finished model checkpoints and source rotation-pretraining weights.
- Test predictions with indices, experiment JSON, per-class metrics and all plotted figures.
- Interactive static GitHub Pages dashboard, 12 real explanation cases, six actual Chromium screenshots.
- Streamlit app for digit image uploads and model inference, plus automated tests and publication workflow.

What this benchmark cannot demonstrate

- The source pictures are just 8x8 grayscale digits; upscaling to 32x32 does not create visual detail.
- Within-domain rotation pretraining is genuine source-task transfer but it is NOT ImageNet pretraining.
- Only one random split/seed was executed, with unequal pretraining compute and 315 held-out images.
- The ImageNet-to-CIFAR-10 script is implemented but NOT run; no CIFAR-10 metrics are claimed.
- Grad-CAM is an attribution heuristic; its visual plausibility is not evidence of causal explanation.

BTU curriculum alignment

- 14186 Image Processing and Computer Vision: filters, learned image representations and visual tasks.
- 14021 Explainable Machine Learning: class attribution and neural visualization.
- 14449 Workflows for Machine Learning and Reproducible Science Laboratory: experiment design, evaluation, code audit and documentation.

OFFICIAL CURRICULUM SOURCE

BTU AI M.Sc. module handbook, summer semester 2026 (pp. 97, 121, 152). See project docs/BTU_ALIGNMENT.md for full links. Curriculum overlap does not guarantee admission.